

Load Profiling and Its Application to Demand Response: A Review

Yi Wang, Qixin Chen, Chongqing Kang*, Mingming Zhang, Ke Wang, and Yun Zhao

Abstract: The smart grid has been revolutionizing electrical generation and consumption through a two-way flow of power and information. As an important information source from the demand side, Advanced Metering Infrastructure (AMI) has gained increasing popularity all over the world. By making full use of the data gathered by AMI, stakeholders of the electrical industry can have a better understanding of electrical consumption behavior. This is a significant strategy to improve operation efficiency and enhance power grid reliability. To implement this strategy, researchers have explored many data mining techniques for load profiling. This paper performs a state-of-the-art, comprehensive review of these data mining techniques from the perspectives of different technical approaches including direct clustering, indirect clustering, clustering evaluation criteria, and customer segmentation. On this basis, the prospects for implementing load profiling to demand response applications, price-based and incentive-based, are further summarized. Finally, challenges and opportunities of load profiling techniques in future power industry, especially in a demand response world, are discussed.

Key words: load profiling; demand response; data mining; customer segmentation; Advanced Metering Infrastructure (AMI)

1 Introduction

In order to build a more robust and efficient power grid, countries worldwide have made large investment towards developing smart grids in recent years. Different researchers may hold different views on the definition of the smart grid. The U.S. Department of Energy (DOE) suggests that a two-way flow of electricity and information that can be used to monitoring everything from power plants to customer preferences is one of the core connotation of the

smart grid^[1]. In China, both State Grid Company of China and China Southern Power Grid are pursuing a more intelligent and stronger power grid through wide application of information infrastructure^[2].

The data gathered by Advanced Metering Infrastructure (AMI) is an important component of information flow in a smart grid. Recently, AMI has gained increasing popularity worldwide. According to statistics released by the Institution for Electric Efficiency, USA had installed about 36 million smart meters by May 2012^[3]. It is estimated that nearly 80 million smart meters will be installed in China^[4].

Countries around the world spare no efforts in promoting deregulation of the electrical industry. The Load Servicing Entities (LSEs) will emerge in large numbers and form a more competitive electrical market, especially on the demand side, thus challenging the current distribution and supply structures. In this context, LSEs would be needed to make full use of detailed electrical power consumption data of individual customer obtained by AMI to gain a better

• Yi Wang, Qixin Chen, and Chongqing Kang are with Department of Electrical Engineering, Tsinghua University, Beijing 100084, China. E-mail: wangyi14@mails.tsinghua.edu.cn; qxchen@tsinghua.edu.cn; cqkang@tsinghua.edu.cn.

• Mingming Zhang, Ke Wang, and Yun Zhao are with Electric Power Research Institute, China Southern Power Grid Co., Ltd., Guangzhou 510080, China.

* To whom correspondence should be addressed.

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understanding of consumer behaviors^[5]. Traditionally, customers are roughly classified as residential, commercial, industrial, and so on. However, often, consumers of the same type have completely different electrical consumption patterns. To address this problem, load profiling, which refers to an electricity consumption pattern for a customer over a given period, is performed. This help both LSEs and electrical customers enhance the understanding of electrical consumption patterns for realizing personalized power management and activating the interaction between LSE and electrical customers in a competitive electric power retail market.

Generally, as shown in Fig. 1, the load profiling process can be divided into five stages: load data preparation, load curve clustering, clustering evaluation, customer segmentation, and result application.

Stage 1: Load data preparation. The electrical consumption data gathered by smart meters of individual users may contain bad data. Data clearing, which will not be discussed in this paper, should be conducted first.

Stage 2: Load curve clustering. Clustering, an unsupervised classification technique, separates enormous load curves into several clusters. The center of each cluster is a typical load curve. Clustering is the core technique of load profiling. A variety of clustering methods have been explored by researchers.

Stage 3: Clustering evaluation. Clustering should make the load curves in the same cluster share similar patterns, while the curves in different clusters should show significant difference. Various criteria can be employed to evaluate clustering performance by quantifying these similarities and differences. These criteria can be used to choose the number of clusters and compare clustering techniques with each other.

Stage 4: Customer segmentation. The process of customer classification is designed to assign a new consumer to a particular customer group by recognizing the customer's load profile pattern. The classification procedure is very important because we can apply the classification rule to judge the ascription of a newcomer with little information or mine additional information from the load curves.

Stage 5: Result application. Load profiling has many applications including demand response, load forecasting, and non-technical loss detecting, etc. Customers' active participation in the smart grid is a crucial element of the two-way power flow, and this can be achieved via the notion of demand response. Demand response, a very active area of research all over the world, is an effective way of promoting the accommodation of renewable energy and reducing the difference between peaks and valleys of electrical load^[6, 7]. The electricity consumption data and load profiles extracted from these data are vital for demand response studies. In this study, we focus on the prospects of implementing load profiling in demand response applications.

Several review works on load profiling have been published^[8-12]. Prahastono et al.^[8] gave an introduction to the five most popular clustering techniques including hierarchical clustering, k -means, follow-the-leader, fuzzy k -means, and fuzzy classification, and then analyzed the difference among the methods. In Ref. [9], frequently used clustering techniques were reviewed and their performance was assessed. A detailed comparison of these proposed techniques was conducted. Similar work has been done in Ref. [12]. Zhou et al.^[11] also summarized briefly three clustering techniques, namely, k -means, fuzzy k -means, and hierarchical clustering. They discussed the application of load profiling on bad data identification, load forecasting, and tariff setting. While, McLoughlin et al.^[10] discussed some time series methods like Fourier transforms and Gaussian processes. It can be seen that most of them are focused on the various clustering technologies and its evaluation of performance. However, the focus of this paper is different. The main contributions of this paper include: (1) extending some new reported clustering techniques especially for the indirect clustering to present a more comprehensive summary; (2) proposing a review on the classification techniques of customers which is very important for customer segmentation; (3) focusing on the application of load profiling to demand response; (4) addressing some challenges and opportunities of load profiling in the big data and Energy Internet era.

The rest of this paper is structured as follows.

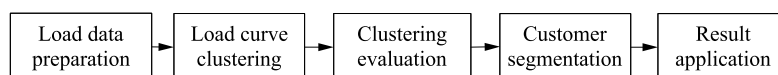


Fig. 1 Basic procedures of load profiling.

Section 2 provides the state-of-the-art review on the clustering techniques for load profiling. The evaluation criteria of the performance of clustering techniques are summarized in Section 3. Section 4 gives a summary on the classification methods for customer segmentation. The application of load profiling to price-based and incentive-based demand response is described in Section 5. Section 6 addresses the challenges and opportunities of load profiling. Finally, the conclusions are drawn in Section 7.

2 Clustering Techniques for Load Profiling

2.1 An overview of clustering techniques

The clustering techniques in this paper are classified into two categories: direct clustering and indirect clustering. We define direct clustering as a clustering method on the data collected by smart meters directly. While, if the load data has been processed by dimension reduction techniques or other method before clustering, we classify it as indirect clustering. Figure 2 gives the detail classification of clustering methods which are summarized in Subsections 2.2 and 2.3 respectively.

2.2 Direct clustering

On the basis of load data collection and processing, various clustering techniques are used to detect the electricity consumption patterns. The detail procedures, pros and cons of the most frequency used techniques, including k -means, fuzzy k -means, hierarchical clustering, and Self-Organizing Map (SOM), have been provided and analyzed in different papers^[11–13], which will not be repeated in the paper. Researchers around the world attempt to look for a better way to perform electricity consumption patterns. These traditional clustering methods are often used as a

benchmark to assess other new reported methods. It should be noted that the clustering validity indicators mentioned below can be referred in Section 3.

Especially, In Ref. [13], five frequently applied clustering methods (k -means, fuzzy k -means, hierarchical clustering, modified follow-the-leader, and SOM) have been tested on 234 non-residential customers connected to medium voltage distribution systems. The clustering algorithms were run 96 times with the number of clusters varying from 5 to 100. Different types of validity indicators have been calculated to evaluate the performance of these methods. The results verify the superiority of the modified follow-the-leader and hierarchical clustering methods, which can handle isolated uncommon load patterns. The modified follow-the-leader and SOM methods are discussed in Ref. [14] as well.

Chicco et al.^[15] adopted an original Electrical Pattern Ant Colony Clustering (EPACC) algorithm to obtain typical electrical load curves. The ant colony system is a hot issue in artificial intelligence, which is often used for solving optimization problems. Essentially, searching for typical load curves is an optimization problem. The authors innovatively introduced the ant colony system to the optimization problem, the objective of which is to minimize two clustering validity indicators: Clustering Dispersion Indicator (CDI) and Scatter Index (SI). In Ref. [16], the authors extended the original Support Vector Clustering (SVC) methodology with a deterministic algorithm to form clusters. SVC maps the data points (load curves) to a high dimensional feature space by employing a Gaussian kernel, and then searches for the minimal enclosing sphere to form a separate cluster of points. The selection of parameters of the Gaussian kernel function directly affects the performance of the clustering method. In addition to achieving better results than other traditional methods, the authors eased part of the procedure to facilitate its application to large number of multidimensional data entries.

Mutanen et al.^[17] proposed an Iterative Self-Organizing Data-Analysis Technique Algorithm (ISODATA) for load profiling considering the dependency of load curves on temperature. The temperature dependency parameters were weighted 5%, and the actual load measurements were weighted 95% in their similarity measure definition. This method iteratively separates an existing cluster into two and merges two existing clusters into one cluster. A case

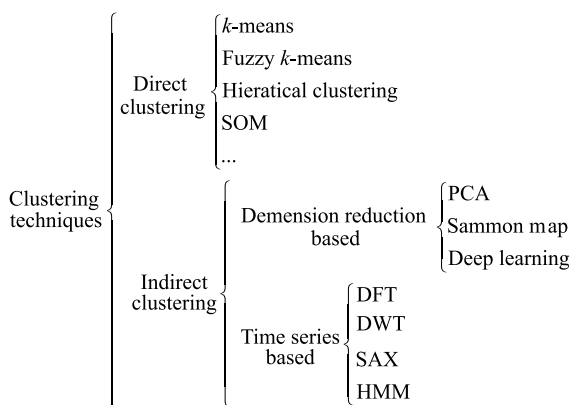


Fig. 2 Classification of clustering techniques for load profiling.

study from Finland showed that the proposed method allows for accurate customers classification.

Ramos et al.^[18] demonstrated an Evidence Accumulation Clustering (EAC) based ensemble combination approach to obtain typical load profiles. Before clustering, the neural net is trained to clear the electricity consumption data. The core of this method is weighing each data partition, where Single-Weighted EAC with Sub-sampling (SWEACS), and Joint-Weighted EAC with Sub-sampling (JWEACS) are used.

Tsekouras et al.^[19] proposed a two-stage technique to classify electrical customers. The typical load curves of individual customers are obtained in the first stage. In the second stage, the objects of clustering are the load curves obtained in the first stage. Both stages employ traditional pattern recognition methods, such as k -means and hierarchical clustering. Six clustering validity indicators were used to evaluate the performance of each method and as guidance to select the best performing method.

The clustering techniques described above are based on the Euclidean distance of all dimensions. This prevents the method from capturing the local shape of a load curve. To address this problem, Piao et al.^[20] introduced subspace clustering load profiling. They tested specific methods belonging to three major subspace clustering approaches (cell-based, density-based, and clustering-oriented) by varying relevant parameters. A case involving 1205 consumers was conducted using k -means, SOM, and different proposed subspace clustering methods. The results reveal that a subspace projection based method named Proclus outperforms the other methods.

2.3 Indirect clustering

As mentioned above, indirect clustering means that the objects of clustering are the features extracted from electricity consumption data instead of the data

itself. Assume that $X_{n \times m}$ stands for an electricity consumption data set of n customers' load data with m dimensions. The procedure of feature extraction is essentially a function of $X_{n \times m}$:

$$Y_{n \times k} = f(X_{n \times m}) \quad (1)$$

where $Y_{n \times k}$ denotes the features extracted from n customers' load data with k dimensions. Feature extraction is often used to reduce the scale of input data, so that the inequality $k < m$ holds generally. Based on the feature extraction techniques employed, the indirect clustering techniques can be categorized into dimension reduction based and time series based clusterings.

2.3.1 Dimension reduction based clustering

Dimension reduction is an important technique to deal with problem of high dimensional data and compress the size of data. Even though, a great deal of work has been done on this field of linear and nonlinear manifold learning, the dimension reduction issue is still a hot problem in machine learning. We have reasons to believe that the daily electricity consumption of each is interrelated and can be represented with a smaller set of artificial variables.

Principle Component Analysis (PCA), a typical linear dimension reduction method, tries to retain most covariance of the data features with the least artificial variables. The principle of PCA is shown schematically in Fig. 3a. In Ref. [21], before fuzzy k -means clustering, the authors applied PCA to discover the most various or valuable features of the daily electricity consumption data. The example shows that three components can cover over 95% of the covariance of a daily load curve with 48 values. A similar two-stage method was used by Koivisto et al.^[22] to model the electricity consumption of connection points, which means the total consumption of individual customers. The authors pointed out that PCA can help us gain good knowledge of consumption behaviors, because they found that some coefficients of

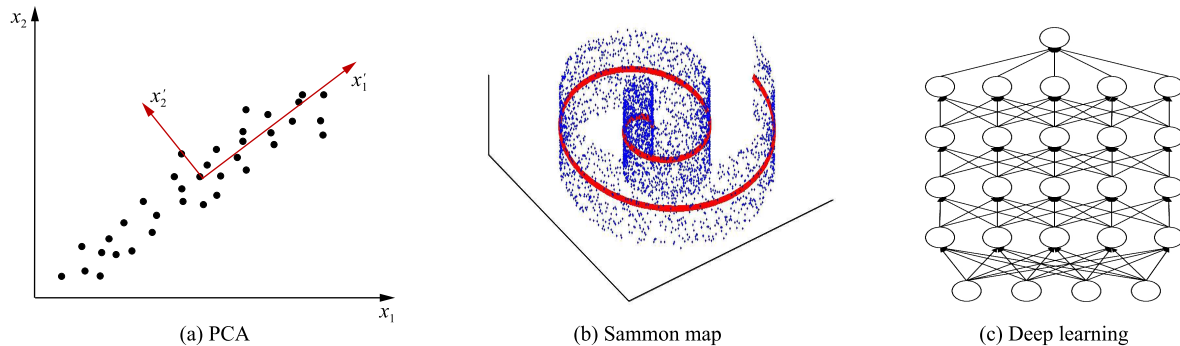


Fig. 3 Principle of dimension reduction techniques.

components are correlated with temperature.

Sammon map is a nonlinear dimension reduction method that can preserve the metric relations of the data items, as shown in Fig. 3b. In Ref. [23], both Sammon map and another method called Curvilinear Component Analysis (CCA) were tested on non-residential customers. The validity indicator, Davies Bouldin index (DBI), suggests that Sammon map performs better than CCA and PCA.

Deep learning, which launched the second wave of machine learning, was developed by Geoffrey Hinton and his students in 2006^[24]. Deep learning is essentially a multi-layer neural network, as shown in Fig. 3c. Lower dimensional data can be learnt in the neural network from higher dimensional data. Varga et al.^[25] proposed the use of a deep learning auto-encoder for load profiling in real-time.

Except for the methods summarized above, linear dimension reduction methods, including Projection Pursuit (PP), and nonlinear methods, including Multi-Dimensional Scalar (MDS), Isometric Map (ISOMAP)^[26], and Locally Linear Embedding (LLE)^[27], can be applied as well.

2.3.2 Time series based clustering

Note that electricity consumption data is essentially a time series. A variety of mature time series analytical methods can be applied to extract the features of individual consumers' load data.

Discrete Fourier Transform (DFT) is the frequency domain representation of an original signal consisting of a series of amplitudes and phases of different frequencies. Under certain circumstances, the time series can be represented neatly in the frequency domain. Zhong and Tam^[28] firstly transformed the electricity consumption data into the frequency domain, and then established a Classification And Regression Tree (CART) framework to classify

different consumers. In Ref. [29], both the time series itself and the data in frequency domain are subjected to SOM clustering. The mean quantization error and the topographic error, which represent map resolution and the map distortion respectively, were calculated to quantify the performance of SOM. The result improves when the input data is in the frequency domain.

Discrete Wavelet Transform (DWT) is another time frequency transformation technique. It is called the microscope of mathematics for its ability to transform time series with multi-resolution. Xiao et al.^[30] selected Haar wavelets to model electricity consumption data and then used k -means to obtain typical load curves. However, the lack of comparative results means that the superiority of the proposed method over other methods cannot be verified.

Symbolic Aggregate Approximation (SAX), a dimensionality reduction method for time series, performs as well as DWT and DFT, while requiring less storage space^[31]. SAX maps the time series to a time amplitude as shown in Fig. 4a. In Ref. [32], the authors adopted this method to form a reduced set of features and pointed out that the dimension of the SAX word and the alphabet size influence the features size.

Hidden Markov Model (HMM) is powerful tool for modeling Markov processes with hidden states. As shown in Fig. 4b, both the transition of hidden states and the output of a certain state are stochastic in HMM. Albert and Rajagopal^[33] used a regression model to determine the occupancy-based consumption instead of weather-based assumption assuming that weather and occupancy are independent. Then, an HMM framework was used to determine the magnitude, duration, and variability of occupancy. Finally, spectral clustering was used to aggregate the customers with similar dynamics.

In addition to these four presented methods, other approaches, including Singular Value Decomposition,

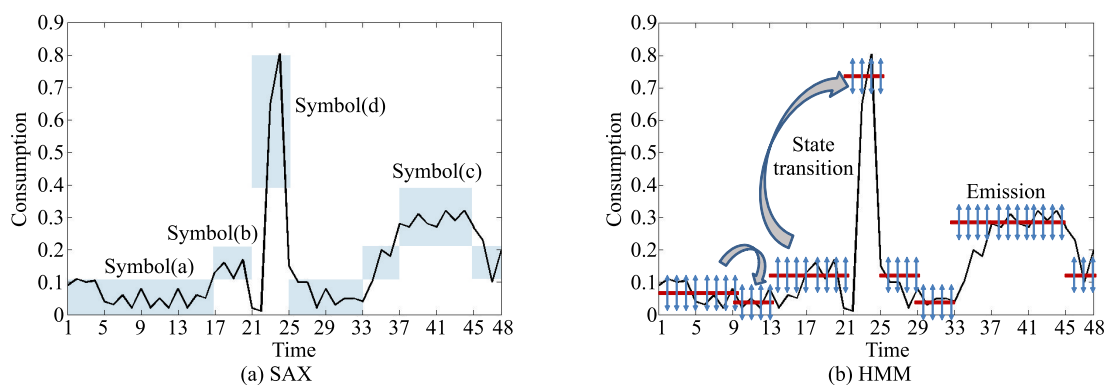


Fig. 4 Principle of SAX and HMM.

Adaptive Piece Wise Constant Approximation, Piecewise Aggregate Approximation (PAA), and Piecewise Linear Approximation, can reduce the dimension of time series as well as maintain the original characteristics of electricity consumption data^[34].

2.4 Discussion of clustering techniques

Researchers' studies on load profiling focus mainly on clustering techniques. The case studies in different papers are various, depending mainly on the types of electricity consumption data collected by the authors and the study purpose. These data differ in terms of voltage levels (low-voltage and medium-voltage) and temporal resolution (15 min, 30 min, and 1 h). Also, when applying clustering techniques, the definition of dissimilarity among different load curves and the number of clusters are diverse. In this part, discussions on voltage levels, temporal resolutions, distance metrics, and number of clusters are presented.

Voltage levels. In various studies, the objects of load profiling differ across various consumer groups such as residential customers and medium-voltage customers. Residential customers have greater variation in load usage patterns, while medium-voltage customers show greater regularity. Clustering techniques may show different performances in different situations.

Data resolutions. The data resolution depends mainly on the performance or settings of smart meters. Studies on load curves have dealt mostly with three kinds of temporal resolutions: 15 min, 30 min, and 1 h. Granell et al.^[35] studied the influence of temporal resolution of load curves on the resulting clusters. Given that smart meters are equipped with the ability to collect high resolution data, the effect of load curve resolution on the quality of clustering should be studied carefully. The authors examined the clustering quality and cluster membership consistency of k -means and Dirichlet Process Mixture Model (DPMM) algorithms by varying the resolution of the raw data. The result shows that a frequency of at least 30 min can secure high-quality clustering, while 60 min is not reliable.

Distance metrics. The metric for similarity between two objects is a fundamental issue in clustering. Substantially, the similarity among electrical load curves is quantified by Euclidean distance, as expressed by Eq. (2).

$$d(x, y) = \sqrt{\sum_{h=1}^H (x_h - y_h)^2} \quad (2)$$

Very few papers define new metrics of similarity. In Refs. [36, 37], a weighted Euclidean distance was proposed to classify similar load curves into one cluster, according to Eq. (3).

$$d(x, y) = \sqrt{\sum_{h=1}^H \frac{\sigma_h^2}{H\bar{\sigma}^2} (x_h - y_h)^2} \quad (3)$$

where σ_h^2 stands for the variance of the h -th value of all load curves and $\bar{\sigma}^2$ stands for the average value of all H variance. It can be seen that features with bigger variation weigh higher.

Now that the load curve is a time series, many distance measures can be applied. The Mikowski distance (a generalization of Euclidean distance) focuses more on the shape of load. Dynamic time warping distance is a good tool for discovering load usage patterns that can weaken the corresponding relationship with time. Kullback-Liebler distance can be used to measure the difference between transition probabilities of two Markov Chains (MCs), which can find the load curves with similar dynamic characteristics.

Number of clustering. The selection of cluster number needs an all-encompassing consideration. On the one hand, clustering performance varies with the number of clusters. Generally, clustering should be run many times to choose the number of cluster that yields the best performance indicators. Panapakidis et al.^[38] used the Ratio of Within Cluster sum of squares to Between Cluster variation (WCBCR) as an adequacy measure and the Bayesian Information Criterion (BIC) to evaluate algorithm effectiveness. Then they applied a knee point detection method to WCBCR in order to choose the optimal number of clusters. On the other hand, there are requirements pertaining to the clusters number in the load profiling applications such as tariff design. Result with too many clusters makes the tariff design issue complex. While too few clusters do not provide sufficient information about electrical customers.

3 Evaluation Criteria

On the basis of clustering results, various evaluation criteria can be used to quantify the performance of different clustering techniques and lead us to choose an appropriate number of clusters. The evaluation criteria can be categorized mainly as similarity-oriented and classification-oriented.

3.1 Similarity-oriented criteria

Clustering is used to aggregate similar members into a cluster and separate members with significant differences into different clusters. A good clustering technique ensures that each cluster is compact, and that various clusters are well widely separated from each other. Various validity indicators have been proposed to quantify the performance of different clustering techniques. Table 1 lists the four most widely used validity indicators. Chicco^[9] summarized other validity indicators, including the Intra-cluster Index (IAI), Modified Dunn Index (MDI), Variance Ratio Criterion (VRC), and Inter-cluster Index (IEI).

Although these validity indicators are used to characterize the result of clustering, there may be little difference among them. In Ref. [43], the authors tested two clustering methods, Weighted Fuzzy Average (WFA) k -means and modified follow-the-leader, by MIA and CDI, which show the separation and compactness of clusters, respectively. The case study in that paper shows that WFA k -means has a lower MIA, while modified follow-the-leader has a lower CDI. The authors believed that WFA k -means is suitable for classification when the inter-cluster difference is large, such as in the case of tariff design, while modified follow-the-leader can be applied to classification issues where the patterns in each cluster exhibit similarity, such as in the case of the demand response issue.

3.2 Classification-oriented criteria

If there are labels for individual customers, such as residential, commercial, and industrial, a good clustering technique should aggregate customers with the same label into a cluster and separate customers with different labels into different clusters. For dualistic classification, we define TP, FN, FP, and TN as the number of positives correctly predicted

as positives, positives wrongly predicted as negatives, negatives wrongly predicted as positives, and negatives correctly predicted as negatives, respectively. We can calculate the accuracy, precision, recall, and F_1 of the classification technique using Eqs. (4)–(7). The higher these four indexes, the better is the classification technique. Piao et al.^[20] applied these criteria to verify whether subspace clustering can separate customers with higher accuracy. Granell et al.^[35], too, studied the effect of temporal resolution on the quality of clustering by using these four indicators.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

$$F_1 = \frac{2}{1/\text{Precision} + 1/\text{Recall}} \quad (7)$$

3.3 Other criteria

Zhang et al.^[44] provided a new index for selecting a stable clustering algorithm. They found that multiple runs of a certain clustering technique on certain clusters may yield different results. If a certain clustering technique has been run T times, the stability index is defined as follows:

$$S = \frac{G}{N} \quad (8)$$

where G stands for the number of load pattern groups of T -run and N stands for the given number of clusters. The stability index can reflect the stability of clustering techniques. The smaller the stability index, the better is the clustering technique.

4 Classification Methods

Classification, a supervised data mining technique, is used to train a target function to map an attribute set

Table 1 Similarity-oriented criteria.

Indicator	Formula	References
Clustering Dispersion Indicator (CDI)	$CDI = \frac{1}{\hat{d}} \sqrt{\frac{1}{K} \sum_{k=1}^K \hat{d}^2(X^{(k)})}$	[14,16,19,23,37,39,40]
Scatter Index (SI)	$SI = \left(\sum_{m=1}^M d^2(x^{(m)}, p) \right) \left(\sum_{k=1}^K d^2(c^{(k)}, p) \right)$	[16,23,39,40]
Davies Bouldin Index (DBI)	$DBI = \frac{1}{K} \sum_{k=1}^K \max_{i \neq j} \frac{\hat{d}(X^{(i)}) + \hat{d}(X^{(j)})}{d(c^{(i)}, c^{(j)})}$	[19,23,39,40,41]
Mean Index Adequacy (MIA)	$MIA = \sqrt{\frac{1}{K} \sum_{k=1}^K d^2(c^{(k)}, X^{(k)})}$	[14,16,19,23,37,39,40,42]

to a predefined class label. The classification in load profiling is modeled for assigning unknown consumers into existing classes.

Decision tree uses a tree-like model consisting of root nodes, internal nodes, and leaf nodes to support a decision. Once the decision tree has been structured, the classification will become very easy. Chang and Lu^[45] proposed a C4.5 algorithm for load profile class assignment. The attribute set of the decision tree consists of customer monthly energy usage and change data. The C5.0 algorithm, a more robust and fast method, was applied in Ref. [46]. Three load shape indexes (load factor, night impact, and lunch impact) were computed for individual customers as attributes. The quality of C5.0 was evaluated by ten-fold cross validation. Similar to Ref. [32], Ramos and Vale^[47] dealt with medium voltage customers using C5.0, the attributes of which are five load shape indexes.

Multinomial logistic regression extends classical logistic regression to multiclass problems. It can be used only when the attributes are nominal. Li et al.^[48,49] developed a three-stage method to obtain low voltage network load profiles. On the basis of clustering, a multinomial logistic regression was designed for the assignment of unmonitored low voltage substations. The attributes of the proposed regression scheme include customer number in each class, transformer type, and number of feeders instead of the feature extracted from load curve. McLoughlin et al.^[50] selected dwelling, occupants, and appliance characteristic sample sizes as inputs to the regression model in order to capture the characteristics of domestic electricity load profiles.

It should be emphasized that proper selection of attribute set is crucial for ensuring good classification performance. Various attributes can be selected, but the use of too many low-related attributes may result in over-fitting and low-accuracy predictions. References [45–47] selected those features extracted from load curves that are related directly to load shapes, while Refs. [48–50] chose the components of domestic customers or low voltage stations.

5 Application to Demand Response

In deregulated electrical markets, demand response has attracted more and more attention from LSEs and electrical customers. As shown in Fig. 5, demand response can be classified into two categories: price-based programs and incentive-based programs. In price-

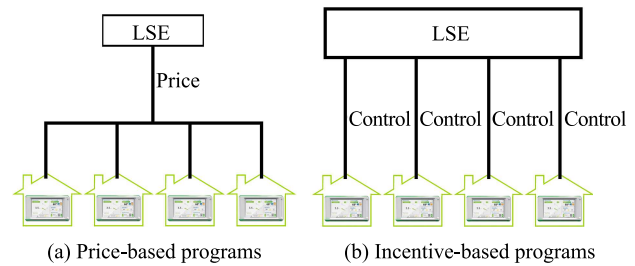


Fig. 5 Price-based and incentive-based demand response programs.

based programs, electrical customers change their electricity usage patterns in response to change in the price of power. Unlike price-based programs, in incentive-based programs, control equipment is installed at customer premises. Using this equipment, LSEs can indirectly switch certain electric appliances at certain times.

For these participants, spending cuts can be achieved by reducing their electrical consumption during peak hours in price-based programs or via incentive payments in incentive-based programs. LSEs can benefit by gathering demand response resources to bid for ancillary services in the markets. More importantly, owing to active participation in the power balance, demand response contributes to the reliability and stability of the entire power system. In addition, it helps avoid or defer the construction or enforcement of distribution and transmission infrastructure^[51]. Load profiling plays a vital role in the design of profitable demand programs by providing necessary information about individual customers, such as their electrical consumption patterns.

5.1 Price-based programs

In price-based programs, customers adjust their electricity consumption in response to the price designed by LSE, including Time Of Use (TOU), Critical Peak Pricing (CPP), and Real Time Pricing (RTP). Without smart meters, only the total electricity consumption of a day, month, or even longer periods is provided to the LSE. Consumers with the same total consumption and different peak consumption are charged the same. However, with the application of smart meters, load profiling can be done to assist the LSE design time-variant prices in order to maximize profit and promote demand response.

The main methodology of price design involves establishing and solving an optimization model on the basis of load profiling. Mahmoudi-Kohan et al.^[52]

proposed a three-stage strategy for TOU design. Following determination of the customers willing to purchase electricity, different prices are optimized for customers belonging to different clusters separately. A simulation involving 300 customers showed that the clustering technique with lower CDI can achieve greater profit. In Ref. [53], Conditional Value at Risk (CVaR) was taken into consideration. The same clustering technique and acceptance function as in Ref. [52] were applied to the stochastic programming model. Load profiling was used as well to find customers with similar peaks in the overall curve, as in Ref. [36]. A cost minimization model is formulated to give priority to customers with high elasticity. The constraint specifies that total load reduction be equal to or higher than the retailer's power shortage. Panapakidis et al.^[54] extracted typical load curves via load profiling and then designed the price for each typical curve. A comparison between annual load profiling and seasonal load profiling was conducted for an industrial consumer. The results showed that profits can be increased by considering the influence of seasonal variation of prices in the pool market. Instead of price design, Maigha and Crow^[55] proposed load profiling-based optimal TOU structures. Degree of cluster granularity, which refers to a scenario where clusters contain less than two hours and those clusters are distributed throughout the day, acts as a performance metric for clustering techniques. The results of a case study verified that Gaussian mixture models performed the best among the compared methods.

5.2 Incentive-based programs

It is costly to enroll a customer into incentive-based programs owing to the installation cost of control equipment and the need to make incentive payments to customers. Targeting suitable electrical customers is an antecedent of incentive-based demand response. This part will provide some examples of customer selection via overall analysis of load profiling.

Kwac et al.^[56, 57] established a load shape dictionary for segmenting residential customers by consumption. Adaptive k -means was used to find representative load shapes, and then hierarchical clustering was applied to summarize the adaptive k -means clustering result for load shape dictionary reduction. Each load curve of each customer matched a representative load shape in the dictionary. An entropy analysis was done for each household using electrical consumption data

of a full calendar year to evaluate the variability of each customer. The shape of each typical load curve was analyzed separately. The authors believed that customers with low entropy and large relative peaks during peak hours have greater potential for demand response programs. It is an effective way of customer targeting for demand response because only consumption data is used.

Moreover, for the segmentation of electrical customers, Albert and Rajagopal^[58] focused on the temperature-sensitivity of residential electricity consumption. The thermal regimes of each individual customer were modeled by a new probabilistic graphical model, wherein each hidden state represented the use of heating or cooling appliances. The customers were classified based on effective duration and effective thermal response (sensitivity). Thermal profiling of customers is very important for demand response because of the great potential for the use of heating or cooling appliances.

Contrary to the reduction of scale of load consumption data, Labeeuw et al.^[59] proposed a data upscaling technique for time series to quantify the potential for demand reduction with wet appliances. Complete and detailed appliance data instead of total consumption is unavailable. To address this problem, a distribution-based clustering technique was applied to a data set with limited project data. Then, Expectation Maximization (EM) clustering was used to obtain typical consumption patterns. Customers with high potential demand response were selected based on their attitudes and appliance profiling.

5.3 Discussion

Owing to the diversity of demand response programs, the application of load profiling to demand response varies. Different researchers have analyzed the problem from different perspectives. It is easy to see that price-based programs are usually formulated as optimization problems, where the object functions and constraints are different from each other. In incentive-based programs, too, the methods and criteria used for evaluating demand response potential are quite different. Entropy analysis, thermal sensitivity analysis, etc., have been proposed to this end. It is clear that electrical consumption data is no longer adequate for accurate targeting. Data of temperature, questionnaires, and even appliance consumption can be used to achieve better results.

6 Challenges and Opportunities

6.1 Big data concern

In the age of big data, load profiling has to deal with various large-volume data. On the one hand, various factors should be considered for gaining a comprehensive understanding of the influence of these factors on electrical consumption. For example, electrical consumption is closely related to meteorology (temperature, humidity, etc.) and date (workday or weekends, summer or winter). Even the economy has an impact on electricity consumption, especially for commercial and industrial customers. However, the majority of the studies on this topic use only electrical consumption data to discover the usage patterns of individual customers. The current clustering objects are load curves, where the distance between two load curves is simple. However, when considering other factors such as temperature, the distance between two load curves must be defined carefully.

On the other hand, when the scale of electrical consumption data skyrockets owing to frequent data collection and population-level customers, its storage and transmission should be considered carefully in the big data age^[60, 61]. We note that the case studies in the review papers contain no more than 10 000 customers. In practical applications, though, there may be millions of customers to be analyzed. We need to judge whether the existing data mining techniques are capable of load profiling for such large numbers of customers. In addition, the efficiency of clustering techniques should be examined. The techniques of cloud computing and distributed computing are in nascent stages of development, and their widespread adoption and use would bring forth the second spring for load profiling.

6.2 Energy internet issue

Renewable energy is integrated widely into power systems and Internet technologies are widely applied. Against that backdrop, the concept of Energy Internet is proposed^[62, 63]. Renewable energy, storage, electric vehicles, and routers are key components of Energy Internet, which also creates new challenges for load profiling in two respects. One is the complexity of customers' electrical consumption data because they may use the electricity generated by their own renewable energy or storage, which makes it difficult to measure precisely their electrical consumption patterns.

The second is brought about by the integration of electrical networks, traffic networks, natural gas networks, and heat supply networks, which changes the traditional electricity consumption patterns of customers. Therefore, customer behavior mining is no longer limited to electrical consumption. It should contain a thorough analysis of energy consumption of customers instead of electrical consumption.

7 Conclusions

With widespread use of AMI equipment, a lot of information can be mined from electrical consumption data of individual customers through load profiling. Load profiling enhances our knowledge of electrical consumption patterns. In this paper, we provided a comprehensive review of data mining techniques for load profiling in terms of clustering techniques and evaluation criteria for clustering results. We tried to emphasize on new reported clustering techniques, especially indirect clustering techniques, and classification procedures. The information embedded in load profiling has great potential for exploitation. Especially, we can investigate further the application of load profiling to demand response, which is categorized into price-based and incentive-based programs. Finally, we discussed some challenges and opportunities of load profiling in the context of big data and Energy Internet.

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Yi Wang received his bachelor degree from Huazhong University of Science and Technology in China in 2014. He is currently working towards the PhD degree in Tsinghua University. His research interests include wind power planning, demand response, and big data application in power system.



Ke Wang received his PhD degree from Tsinghua University in 2011. He is currently a researcher of SEPRI. His research interests include demand response, energy efficiency management, and big data application in power system.



Qixin Chen received his PhD degree from Tsinghua University in 2010. He is currently an associate professor at Tsinghua University. His research interests include low-carbon electricity, power system economics and optimization, and power markets and power generation expansion planning.



Yun Zhao received his PhD degree from Wuhan University, China in 2013. He is currently a researcher of SEPRI. His research interests include the information model and communication technology and big data application in power system.



Chongqing Kang received his PhD degree from Tsinghua University in 1997. He is now a professor at the same university. His research interests include low-carbon electricity, power system planning, renewable energy, and power system load forecasting.